

FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

FINAL EXPLANATION: CHEMISTRY GRADE 10 – SUB-STRAND 1.3 THE PERIODIC TABLE: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

You have now finished the Periodic Table unit, starting with the Mystery Element Card Challenge and ending with the Malindi salt case study. Your job is to write a complete, evidence-based explanation that answers the Driving Question: Why do the 118 elements behave so differently, and how can knowing an element's position in the periodic table help us predict what it will do? Use your Driving Question Board, your Model Versions 1 to the final poster, your trend tables, dot-and-cross diagrams and balanced equations. Write in complete sentences, use scientific words correctly (group, period, outer-shell electrons, cation, anion, valency, oxidation number), and give specific evidence from the lessons for every claim. Write your answers in the space under each prompt. Remember that explaining WHY matters more than listing facts.

Part 1: The Mystery Element Cards and the History of the Periodic Table

In Lesson 1 your class sorted 20 unlabelled element cards (H to Ca) and produced different groupings. (a) Describe two different ways groups sorted the cards and the puzzle the class could not yet explain, using the pairs Li/Na, F/Cl and He/Ne as evidence. (b) Using Lesson 2, explain how Döbereiner, Newlands, Mendeleev and Moseley each tried to organise the elements, and why ordering by atomic number worked better than ordering by mass. (c) Explain how the scientists' strategies were similar to your own card sorting.

Write your answer here:

Part 2: Electron Arrangement and Position in the Table

Using Lesson 3, (a) write the electron configurations in s/p notation (as you did in class) for Na, Cl, Ca, K and Ar. (b) Explain what the group number and the period number tell you about an atom. (c) Use this to explain why Na and K behave alike, and why Cl (Gigiri water treatment) and Ar (welding) behave so differently even though they are neighbours.

Write your answer here:

Part 3: Chemical Families and Trends

Using Lesson 4, (a) name the five chemical families you studied and give the group of each. (b) Describe the trend in reactivity down Group 1 (Li, Na, K) and down Group 17 (F, Cl, Br), using evidence from your Trend Detective Table. (c) Write a Claim-Evidence-Reasoning statement about how the reactivity of potassium compares with that of lithium, using only your table evidence and the idea that position predicts behaviour. Then say what your model could and could not yet explain at that point in the unit.

Write your answer here:

Part 4: From Electrons to Ions, Valency and Formulae

Using Lessons 5 to 7, (a) use dot-and-cross ideas to explain how Na and Cl form ions and why the ions are different. (b) State the oxidation numbers for Groups 1, 2, 16 and 17, and explain why iron at Kisumu Port can show Fe^{2+} and Fe^{3+} . (c) Use the crossover method to derive the formulae of calcium chloride and sodium sulfate (radical SO_4^{2-}), and explain how you can check that each formula is correct.

Write your answer here:

Part 5: Balanced Equations and Answering the Driving Question

Using Lessons 8 to 10, (a) write and balance the equation for sodium reacting with water (products sodium hydroxide and hydrogen) and explain why you change coefficients and not subscripts. (b) Using sodium chloride from the Malindi salt flats, explain step by step how an element's position in the periodic table lets you predict its behaviour, its compound and its equation. (c) State one question that is still open and which sub-strand it leads to.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining the pattern with electron arrangement	Clearly links group number to outer-shell electrons and period number to shells, and uses this to explain why same-group elements behave alike and neighbours differ, with correct examples such as Na/K and Cl/Ar.	States that group number relates to outer electrons and gives a mostly correct explanation, with limited examples.	Describes patterns on the table without linking to electron arrangement, or confuses groups and periods.
Use of evidence from the lessons	Uses specific evidence from several lessons (card sorting pairs, history, Trend Detective Table, dot-and-cross diagrams, balanced equations) and Kenyan contexts to support each claim.	Uses some specific evidence but is general in places or draws on only a few lessons.	Gives few or no specific pieces of evidence, or evidence does not match the claim.
Scientific accuracy of ions, valency and formulae	Correctly writes electron arrangements, ions, oxidation numbers, crossover formulae and a balanced equation, and checks charge balance and atom counts.	Mostly correct with one or two minor errors in notation, formulae or balancing.	Several errors, for example wrong ion charges, wrong formulae, or changing subscripts when balancing.
Reasoning and connection to the Driving Question	Builds a clear chain from position in the table to electrons, ions, valency, formula and equation, and uses it to resolve the Lesson 1 mystery and name an open question for Sub-Strand 1.4.	Links most steps in the chain but with some gaps in reasoning.	Lists facts without connecting them or without returning to the Driving Question.
Scientific vocabulary and communication	Uses terms such as group, period, valence electrons, cation, anion and oxidation number precisely, in clear complete sentences.	Uses most terms correctly with occasional unclear sentences.	Uses terms incorrectly or writes in fragments that are hard to follow.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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